

P4 — Design-Aware Main Estimate (Lab 5)

Model submission — demonstration paper (Kueh et al., BMJ Open 2026).

Estimand convention, used identically in every increment.

- **Scientific (locked) estimand — IV vs III.** Non-obese with high central adiposity against non-obese with low central adiposity. This isolates central adiposity with body-mass category held fixed, and it is the question the term is built on.
- **Replication estimand — IV vs I.** The paper’s own parameterization, kept because reproducing it is what *Replicate* means.

Both are contrasts within one fitted model. Every estimate below states which it is.

The estimand, stated before the model

We fix the target of inference first, because “the association” is under-specified until we name a population. Our contrast is the all-cause mortality hazard of phenotype **Group IV** (non-obese by BMI, high central adiposity WHtR ≥ 0.6) relative to **Group I** (obese, low central adiposity), among adults who meet FLI-based MASLD criteria, over follow-up censored 31 December 2019 (median 6.7 years). The measure is a hazard ratio. We state it for the replication contrast (IV vs I); the matrix below treats the locked IV-vs-III contrast in the same way.

The design question is *which* population that HR describes. The paper’s unweighted Cox model estimates a **sample-conditional** association inside the people who happened to be sampled (6,371 in our reproduction of its cohort). But NHANES is a stratified, clustered, unequal-probability survey; without weights the estimate is not a statement about US adults with MASLD. Making the estimand *population-average* — the target-of-inference we think the paper actually wants — requires the survey machinery. So P4 is not “add weights and see if the number moves.” It is a change of estimand, and we report both because both are legitimate answers to *different* questions.

Replicate: the unweighted Cox model

We first reproduce the paper’s Table 2 on the full cohort (n=6,371; 586 all-cause deaths), Group I as reference. Group IV is the headline “worst survival” phenotype:

Model	Group II	Group III	Group IV
Unadjusted	6.68 (2.76–16.14)	6.23 (2.52–15.41)	15.14 (6.19–37.04)
Model 1 (clinical)	2.51 (1.03–6.10)	3.24 (1.31–8.04)	3.29 (1.33–8.14)
Model 2 (+ smoking, sedentary)	2.37 (0.97–5.77)	3.00 (1.21–7.45)	2.91 (1.18–7.21)

These are close to the publication without matching every cell. The unadjusted Group IV estimate agrees closely (15.14 against the paper’s 15.13); our adjusted Group IV estimates sit slightly above the paper’s (Model 1 3.29 against 3.155; Model 2 2.91 against 2.893). The fifteen-fold crude hazard falls to roughly three-fold once the Model 1 or Model 2 covariates are added. As P2 sets out, that attenuation cannot by itself tell us how much of the crude signal was confounding: Model 1 contains candidate mediators, and hazard ratios are non-collapsible. Group IV’s Model-2 confidence interval still excludes 1, but after adjustment Groups III and IV are close (Model 2: 3.00 and 2.91).

Improve: a design-aware estimate

The analytic cohort is the NHANES **fasting subsample**, because the FLI needs fasting triglycerides. The correct weight is therefore the pooled fasting-subsample weight, WTSF2YR divided by 6 (six two-year cycles), with SDMVSTRA strata and SDMVPSU clusters. Valid weights exist for **n=6,048 across 184**

clusters. On that same subset the unweighted crude IV-vs-I HR is 16.96 (6.24–46.05), slightly higher than the full-cohort 15.14 — the weight-eligible subset is not identical to the full cohort, which itself matters.

Survey-weighted (WTSAF2YR/6, strata, PSU), crude, all-cause, Group I reference, on the 6,048 rows:

Group	Weighted HR (95% CI)
II	5.38 (1.19–24.39)
III	4.04 (0.89–18.34)
IV	12.40 (2.69–57.15)

Because Group I is a tiny, low-event reference (375 people and 4 deaths on these 6,048 rows), the vs-I contrasts are unstable in either weighting scheme. So we also estimate **weighted direct contrasts** against the other high-risk groups, which is the comparison a clinician actually cares about:

- **IV vs III: 3.07 (2.11–4.46)** — non-obese/high-central vs non-obese/low-central
- **IV vs II: 2.30 (1.76–3.02)** — non-obese/high-central vs obese/high-central

*The table and both contrasts above are **crude** survey-weighted estimates on the 6,048-row locked domain. The matrix below uses the 5,939 complete-case rows, which is why its weighted crude IV vs III has a slightly different interval. The adjusted versions, and the comparison that separates adjustment from design, are in the matrix below.*

The matrix that separates adjustment from design

The running ledger of $15.1 \rightarrow 2.9 \rightarrow 12.4$ mixes crude and adjusted models on different row sets (6,371 rows for the first two, 6,048 for the third), so it is not a sequence of successive corrections and should not be read as one. To separate **adjustment** from **survey design** the rows must be held fixed.

All four cells below use the **same 5,939 Model-1 complete-case rows** inside the locked domain; “weighted” means WTSAF2YR/6 with strata and PSU, the design built on the full fasting frame and then subset to these rows.

	IV vs I (<i>replication</i>)	IV vs III (<i>locked</i>)
unweighted, crude	16.05 (5.91–43.58)	2.34 (1.72–3.18)
unweighted, Model 1	3.81 (1.39–10.43)	1.05 (0.77–1.45)
weighted, crude	11.93 (2.60–54.84)	3.07 (2.11–4.48)
weighted, Model 1	2.98 (0.69–12.82)	1.29 (0.89–1.86)

Read down a column for what **adjustment** does; read across a row for what **design** does.

- **On the locked estimand, adjustment dominates design.** Adjustment moves 2.34 to 1.05 unweighted and 3.07 to 1.29 weighted. Design moves 2.34 to 3.07 crude and 1.05 to 1.29 under Model 1 — in the *opposite* direction, and by less.
- **Design changes no conclusion on the locked estimand.** Both Model-1 intervals include 1 (0.77–1.45 and 0.89–1.86).
- **Design does change the conclusion on the replication estimand.** Unweighted Model 1 is 3.81 (1.39–10.43), excluding 1; weighted Model 1 is 2.98 (0.69–12.82), which does not.
- So “survey design matters” is true **of the paper’s own contrast**, and much weaker for the question we locked. Anyone ranking design as the dominant threat has to say which estimand they mean.

Interpretation

Two things happen when we move from sample-conditional to population-average. Hold the rows and the model fixed (the crude IV-vs-I estimate on the 6,048-row locked domain) and change only the weighting:

unweighted 16.96 (6.24–46.05), weighted 12.40 (2.69–57.15). First, the point estimate **moves**. Weighting changes how much each respondent contributes according to the sampling design, and it has no fixed direction: here the estimate falls, but on the same rows the crude IV-vs-III estimate rises ($2.33 \rightarrow 3.07$), as it does in the matrix above. Second, the confidence interval **widens**. In the course’s checks, most of such a cost comes from the unequal weights, which reduce the *effective* sample size below the nominal number of rows, rather than from clustering: in the Week 6 lecture’s exposure analysis (set L, 5,911 rows), the strata and PSUs alone raise the variance of its waist-fat estimate among the non-obese by about 5%, and the weights alone by about 56%. That check was made on a different estimate in the lecture’s analysis; we have not repeated it for this one. The unweighted interval is not wrong. It is a valid interval for the sample-conditional estimand; the wider weighted interval belongs to the population-average estimand.

What the matrix actually supports. For the **locked IV-vs-III** estimand, adjustment is consequential and survey weighting changes the point estimate modestly (Model 1 on 5,939 rows: $1.05 \rightarrow 1.29$) *without* altering the inferential conclusion — both Model-1 intervals include 1, so there is **no clear evidence of a higher hazard at the available precision**. For the **replication IV-vs-I** estimand, weighting materially changes the uncertainty and whether the interval excludes 1 (3.81, 1.39–10.43 becomes 2.98, 0.69–12.82). Group IV having the largest point estimate describes an *ordering*; it does not establish that central adiposity carries excess mortality among non-obese adults after adjustment.

Which weight: a rule, not a judgment call

An earlier draft of this submission called the weight choice “a genuine puzzle we cannot fully resolve” and argued the MEC weight was equally defensible. **That was wrong, and we withdraw it.**

CDC’s guidance is the **least common denominator rule**: when an analysis draws on variables measured in different components, use the weight belonging to the **smallest** applicable subsample. FLI requires fasting triglycerides, so **WTSF2YR** governs — even though GGT and anthropometry are available MEC-wide. The rule exists precisely because analyses routinely straddle frames; it is not a tie to be broken by argument.

The MEC weight is not a defensible primary here. Both weights refer to the same population, but **WTMEC2YR** applied to fasting rows leaves out the fasting subsample’s own selection and nonresponse adjustment, which **WTSF2YR** carries. The P4 handout asks for a **WTMEC2YR**-weighted run beside the primary, labelled as a sensitivity analysis; this submission does not include one, so that comparison remains open for M1.